Structural Design

# Tree Fort

Report Example

Proj. 0004

## Contents

## 1.1 | Introductionv

This is an example of a rivtbook. rivtbooks are collections of rivt files organized around a common subject. Each rivt file (subdivision) and its associated resources are contained in a unique folder to faciliate direct selection and insertion into rivt docs and reports.

## 1.2 | Compression Shape Factor

COMPRESSION BLOCK [1]

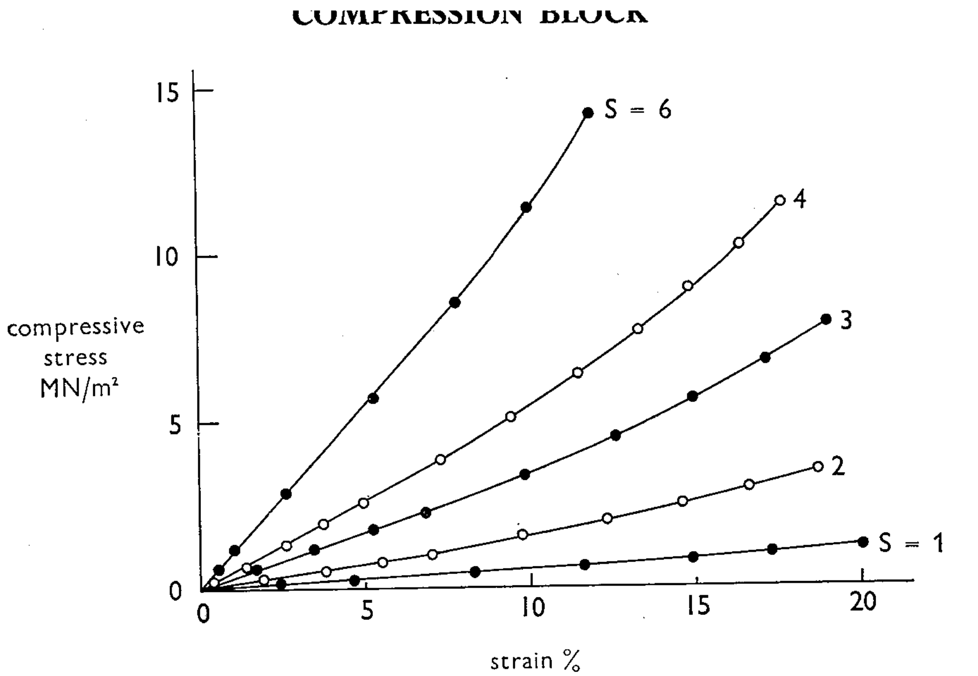


*Fig. 27.—Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.*

**Fig. 1** - Shape Factor

Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.

The stiffness of rubber in compression, when the loaded surfaces are prevented from slipping (by bonding or by mechanical location), depends upon the shape factor S (see Fig. 1), defined as the ratio of one loaded area to the total force-free area, as shown in Fig. 2.

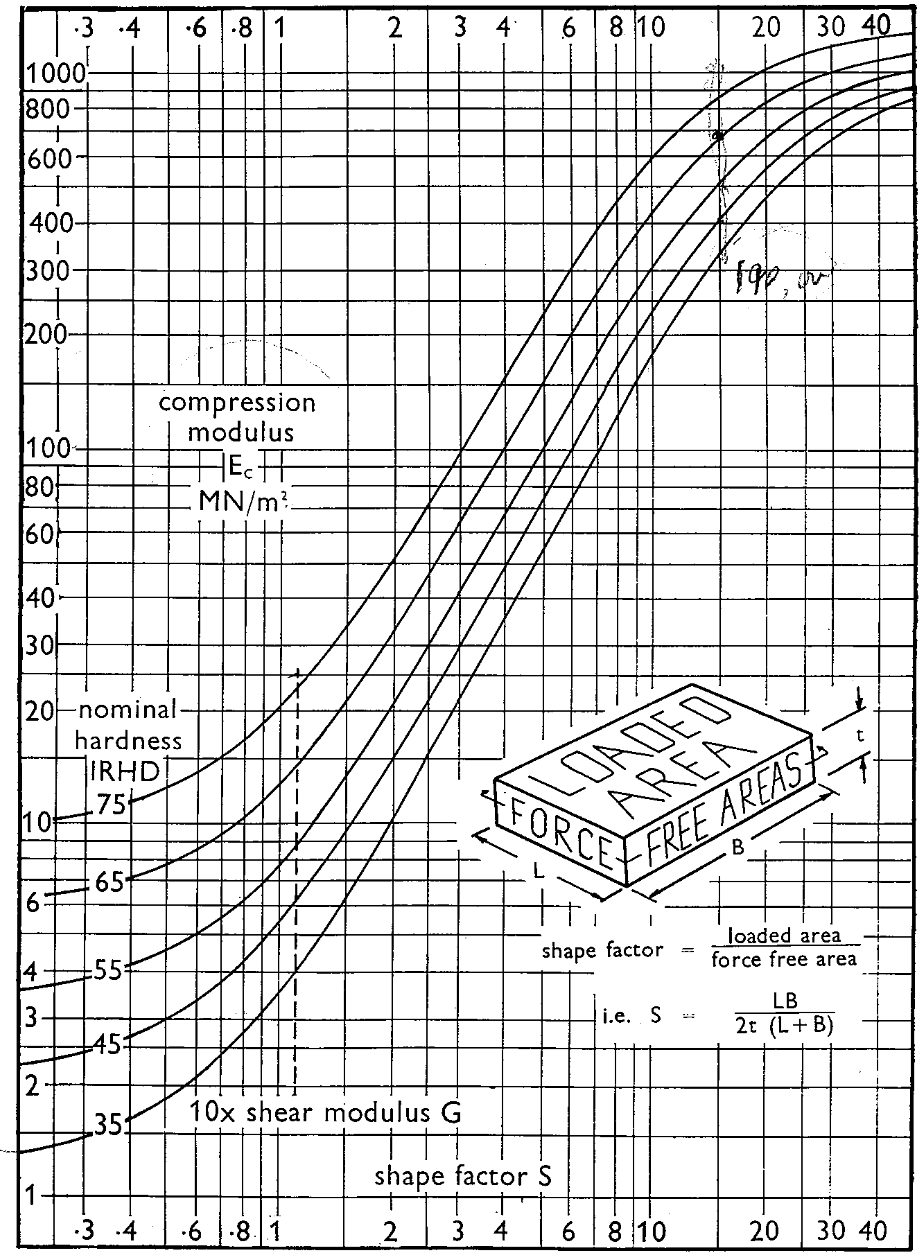


*Fig. 28.—Variation of compression modulus $E_c$ with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above).*

* Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in $E_c$ when the ratio $E_c/E_\infty$ exceeds about 10%. ($E_\infty$ is the modulus of bulk compression, from Table 3.) To allow for this reduction, use a modified compression modulus obtained by dividing $E_c$ by $1+(E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred (see footnote to Table 2) then the $E_\infty$ value in Table 3 should be doubled.

**Fig. 2** - Compression Modulus

Variation of compression modulus Ec with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above). Chart shows compression modulus Ec (MN/m2) on the vertical axis (log scale, 1 to 1000) versus shape factor S on the horizontal axis (log scale, 0.3 to 40), with separate curves for nominal hardness values of 35, 45, 55, 65, and 75 IRHD, and a line labeled “10x shear modulus G”. An inset diagram illustrates a rubber block showing the “loaded area” (top and bottom faces) versus “force free areas” (side faces), with shape factor = loaded area / force free area, i.e. S = (LB) / (2t(L+B)).

Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in Ec when the ratio Ec/E∞ exceeds about 10%. (E∞ is the modulus of bulk compression, from Table 3.) To allow for this reduction, use a modified compression modulus obtained by dividing Ec by 1+(Ec/E∞). If Wood and Martin’s value for bulk modulus is preferred (see footnote to Table 2) then the E∞ value in Table 3 should be doubled.

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## 1.2 - 2 | Compression Modulus

S = (LB) / (2t(L+B))

S = shape factor t = thickness L = length B = breadth

For a block of square section (i.e. L = B) or circular section (diameter = L):

S = L / (4t)

The compression modulus Ec depends upon the shape factor S (for derivation see Appendix).

Ec = E0(1 + 2kS2)

Ec = compression modulus E0 = Young’s modulus (from Table 3) S = shape factor k = a numerical factor (from Table 3)

See footnote on following page.

When S > 3 it may be more convenient to use

Ec ≈ 5GS2

where G is the shear modulus (from Table 3).

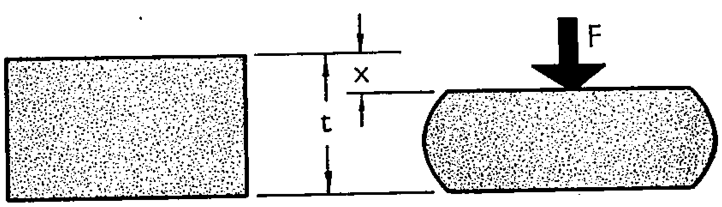


*Fig. 29.—Rubber block in compression.*

**Fig. 3** - Compression Stiffness Kc

Rubber block in compression. Diagram shows a rectangular rubber block of thickness t, deflecting by x under a compressive load F applied to a cylindrical rubber sample.

The compression stiffness, Kc, is given by

Kc = F/x = EcA/t

where F = compressive load Ec = compression modulus (corrected, if necessary, for the effect of bulk compression) A = cross-sectional area t = thickness x = deflexion

It is not advisable to rely on friction alone to prevent slip under a compressive load when using unbonded rubber parts, because slip may occur if S > μ/2, where S is the shape factor and μ the coefficient of friction.

The load (F)—deflexion (x) curve of rubber in compression is non-linear. With no slip it has the approximate form

F = EcAe(1+e)

where e, the compressive strain, equals x/t. The non-linearity is usually ignored for strains up to about 10%.

There is as yet no method of calculating that a block will be stable but experience has shown that provided the thickness is less than one-quarter of the least plan dimension there should be no instability.

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## 1.2 - 3 | Strip Analysis

COMPRESSION STRIP

When a long strip of rubber is compressed the strain in the direction of its length will be negligible.

Shape factor

S = b / (2*t)

IMAGE | img/fig30.png | Compression Strip, 80, num, not

Diagram shows a long rectangular strip of rubber of width b, thickness t, and unit length, compressed by a load F applied per unit length.

Compression modulus Ec

Ec = 4*E0*(1 + k*S**2) / 3

The compression stiffness per unit length, Kc, is given by

Kc = F/x = Ec*b/t = (4*b*E0*(1 + k*S**2)) / (3*t)

where F = load per unit length Ec = compression modulus (corrected, if necessary, for the effect of bulk compression) E0 = Young's modulus (from Table 3) b = width of strip t = thickness of strip x = deflexion k = a numerical factor (from Table 3) S = shape factor

The load per unit length (F)—deflexion (x) curve for a compressed strip is non-linear. It has the approximate form

F = Ec*b*e*(1 + 3*e/2)

where e, the compressive strain, equals x/t. As in the case of blocks, the non-linearity is usually ignored for strains up to about 10%.

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## 1.3 | Rubber Elastic Properties

**Table 1**: IRHD hardness scale (MN/m2)

| Hardness IRHD +/-2 | Young's modulus E, MN/m2 | Shear modulus G, MN/m2 | k | Bulk modulus B(inf), MN/m2 |
|---|---|---|---|---|
| 30 | 0.92 | 0.30 | 0.93 | 1000 |
| 35 | 1.18 | 0.37 | 0.89 | 1000 |
| 40 | 1.50 | 0.45 | 0.85 | 1000 |
| 45 | 1.80 | 0.54 | 0.80 | 1000 |
| 50 | 2.20 | 0.64 | 0.73 | 1030 |
| 55 | 3.25 | 0.81 | 0.64 | 1090 |
| 60 | 4.45 | 1.06 | 0.57 | 1150 |
| 65 | 5.85 | 1.37 | 0.54 | 1210 |
| 70 | 7.35 | 1.73 | 0.53 | 1270 |
| 75 | 9.40 | 2.22 | 0.52 | 1330 |

**Table 2**: Shore A hardness scale (lbf/in2)

| Shore A (approx.) | Young's modulus lbf/in2 | Shear modulus lbf/in2 | k | Bulk modulus lbf/in2 |
|---|---|---|---|---|
| 35 | 168 | 53 | 0.89 | 142 000 |
| 45 | 256 | 76 | 0.80 | 142 000 |
| 55 | 460 | 115 | 0.64 | 154 000 |
| 65 | 830 | 195 | 0.54 | 171 000 |
| 75 | 1340 | 317 | 0.52 | 189 000 |

Note

1. The majority of springs are in the hardness range 40-60 IRHD.
2. Theoretically, with a Poisson's ratio of 1/2, B should equal 3G. This is true for soft gum rubbers. For harder rubbers containing a fair proportion of non-rubber constituents, thixotropic and other effects increase 3G to about 4G.

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## 1.3 - 2 | Shear Stiffness Example

**Table 3**: Bearing Specifications

| variable | value | [value] | description |
|---|---|---|---|
| G_1 | 58.00 lb_in2 | 0.40 MPA | shear modulus |
| K_1 | 200000.00 lb_in2 | 1378.95 MPA | bulk modulus |
| rnl | 51 | 51 | number of rubber layers |
| rdia | 39.50 inch | 100.33 cm | bearing diameter |
| rthk | 0.40 inch | 1.02 cm | layer thickness |

| variable | value | [value] | description |
|---|---|---|---|
| sdia | 38.00 inch | 96.52 cm | shim diameter |
| sthk | 0.09 inch | 0.23 cm | 11 guage shim thickness |
| cpi | 3.1418 | 3.1418 | contant pi |

**Bearing Stiffness for Circular Bearing**

**Eq. 1:** rubber height

```
rht = rnl·rthk

rht = 20.40 inch     [rht] = 51.82 cm   | rubber height
```

| rthk | rnl |
|---|---|
| 0.40 inch | 51 |
| layer thickness | number of rubber layers |

**Eq. 2:** bearing height

```
bht = rht + sthk·(rnl - 1)

bht = 24.95 inch     [bht] = 63.37 cm   | bearing height
```

| sthk | rht | rnl |
|---|---|---|
| 0.09 inch | 20.40 inch | 51 |
| 11 guage shim thickness | rubber height | number of rubber layers |

**Eq. 3:** shear stiffness

$$Ks_1 = \frac{G_1 \cdot cpi \cdot rdia^2}{4 \cdot rht}$$

$Ks_1$ = 3.48 k_i     [$Ks_1$] = 6.10 kN_cm   | shear stiffness

| $G_1$ | rht | rdia | cpi |
|---|---|---|---|
| 58.00 lb_in2 | 20.40 inch | 39.50 inch | 3.1418 |

| shear modulus | rubber height | bearing diameter | contant pi |
|---|---|---|---|

**Shape Factor for Circular Bearing**

**Eq. 4:** shape factor 1

$$sh_{1\ 1} = \frac{sdia}{4 \cdot rthk}$$

$sh_{1\ 1} = 24$ $[sh_{1\ 1}] = 24$ | shape factor 1

| sdia | rthk |
|---|---|
| 38 inch | 0 inch |
| shim diameter | layer thickness |

**Eq. 5:** shape factor 2

$$sh_{2\ 1} = \frac{bht}{rdia}$$

$sh_{2\ 1} = 1$ $[sh_{2\ 1}] = 1$ | shape factor 2

| bht | rdia |
|---|---|
| 25 inch | 40 inch |
| bearing height | bearing diameter |

## 2.1 | Equivalent Viscous Damping

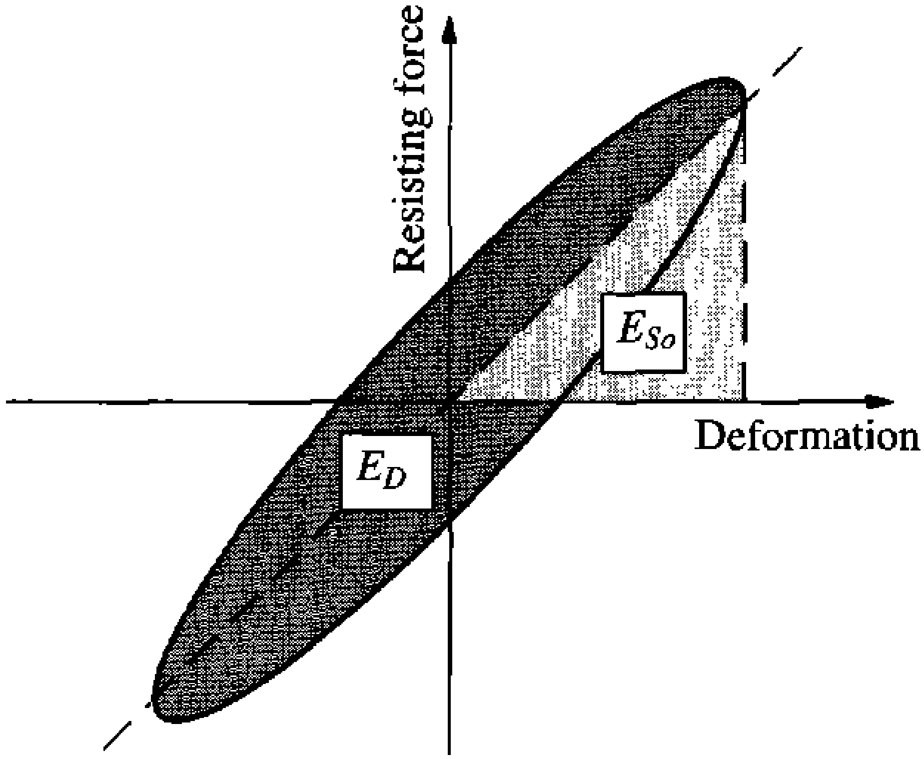


**Fig. 1** - Viscous Damping Model

Damping in actual structures is usually represented by equivalent viscous damping. It is the simplest form of damping to use since the governing differential equation of motion is linear and hence amenable to analytical solution. The advantage of using [2] a linear equation of motion usually outweighs whatever compromises are necessary in the viscous damping approximation. In this section we determine the damping coefficient for viscous damping so that it is equivalent in some sense to the combined effect of all damping mechanisms present in the actual structure; these were mentioned in Section 1.4.

The simplest definition of equivalent viscous damping is based on the measured response of a system to harmonic force at exciting frequency omega equal to the natural frequency omega_n of the system. The damping ratio zeta_eq is calculated from Eq. (3.4.1) using measured values of u_o and (u_st)_o. This is the equivalent viscous damping since it accounts for all the energy-dissipating mechanisms that existed in the experiments.

Another definition of equivalent viscous damping is that it is the amount of damping that provides the same bandwidth in the frequency-response curve as obtained experimentally for an actual system. The damping ratio zeta_eq is calculated from Eq. (3.2.24) using the excitation frequencies f_a, f_b, and f_n (Fig. 3.4.1) obtained from an experimentally determined frequency-response curve.

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### 2.1 - 2 | section-label

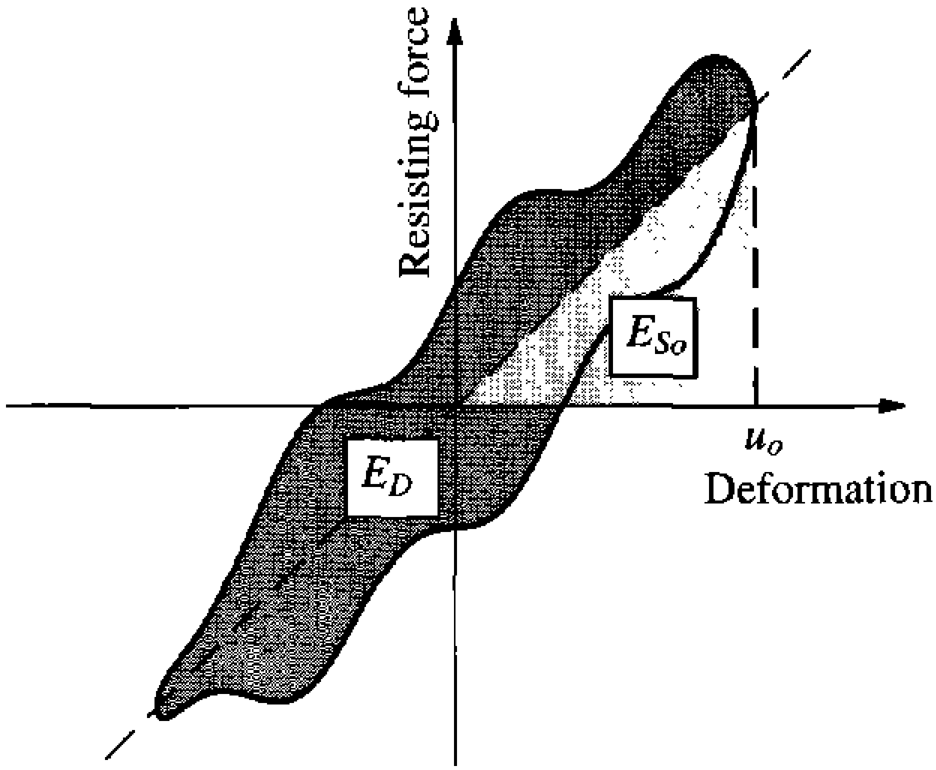


**Fig. 2** - Viscous Damping Model

[Figure 3.9.1 – image file: figures/fig_3_9_1.png – Plot of Resisting force (vertical axis) versus Deformation (horizontal axis), showing an irregularly shaped hysteresis loop determined from experiment. The area E_D is the energy dissipated in a cycle of harmonic vibration, the area E_So is the maximum strain energy, and u_o marks the deformation amplitude on the horizontal axis.]

The most common method for defining equivalent viscous damping is to equate the energy dissipated in a vibration cycle of the actual structure and an equivalent viscous system. For an actual structure the force-displacement relation obtained from an experiment under cyclic loading with displacement amplitude u_o is determined; such a relation of arbitrary shape is shown schematically in Fig. 3.9.1. The energy dissipated in the actual structure is given by the area E_D enclosed by the hysteresis loop. Equating this to the energy dissipated in viscous damping given by Eq. (3.8.1) leads to

4pizeta_eq(omega/omega_n)E_So = E_D

zeta_eq = (1/(4pi)) * (1/(omega/omega_n)) * (E_D/E_So) … (3.9.1)

where the strain energy, E_So = ku_o2/2, is calculated from the stiffness k determined by experiment.

The experiment leading to the force-deformation curve of Fig. 3.9.1 and hence E_D should be conducted at omega = omega_n, where the response of the system is most sensitive to damping. Thus Eq. (3.9.1) specializes to

zeta_eq = (1/(4pi)) * (E_D/E_So) … (3.9.2)

The damping ratio zeta_eq determined from a test at omega = omega_n would not be correct at any other exciting frequency, but it would be a satisfactory approximation (Section 3.10.2).

It is widely accepted that this procedure can be extended to model the damping in systems with many degrees of freedom. An equivalent viscous damping ratio is assigned to each natural vibration mode of the system (defined in Chapter 10) in such a way that the energy dissipated in viscous damping matches the actual energy dissipated in the system when the system vibrates in that mode at its natural frequency.

In this book the concept of equivalent viscous damping is restricted to systems vibrating at amplitudes within the linearly elastic limit of the overall structure. The energy dissipated in inelastic deformations of the structure have also been modeled as equivalent viscous damping in some research studies. This idealization is generally not satisfactory, however, for the large inelastic deformations of structures expected during strong earthquakes. We shall account for these inelastic deformations and the associated energy dissipation by nonlinear force-deformation relations, such as those shown in Fig. 1.3.4 (see Chapters 5 and 7).

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## 2.2 | Hysteretic Damping

3.10 HARMONIC VIBRATION WITH RATE-INDEPENDENT DAMPING [3]

3.10.1 Rate-Independent Damping

Experiments on structural metals indicate that the energy dissipated internally in cyclic straining of the material is essentially independent of the cyclic frequency. Similarly, forced vibration tests on structures indicate that the equivalent viscous damping ratio is roughly the same for all natural modes and frequencies. Thus we refer to this type of damping as rate-independent linear damping. Other terms used for this mechanism of internal damping are structural damping, solid damping, and hysteretic damping. We prefer not to use these terms because the first two are not especially meaningful and the third is ambiguous because hysteresis is a characteristic of all materials or structural systems that dissipate energy.

Rate-independent damping is associated with static hysteresis due to plastic strain, localized plastic deformation, crystal plasticity, and plastic flow in a range of stresses within the apparent elastic limit. On the microscopic scale the inhomogeneity of stress distribution within crystals and stress concentration at crystal boundary intersections produce local stress high enough to cause local plastic strain even though the average “global” stress may be well below the elastic limit. This damping mechanism does not include the energy dissipation in global plastic deformations, which, as mentioned earlier, is handled by a nonlinear relationship between force fS and deformation u.

The simplest device that can be used to represent rate-independent linear damping is to assume that the damping force is proportional to velocity and inversely proportional to frequency:

fD = (ηk/ω) * u̇ … (3.10.1)

where k is the stiffness of the structure and η is a damping coefficient. The energy dissipated by this type of damping in a cycle of vibration at frequency ω is independent of ω (Fig. 3.10.1). It is given by Eq. (3.8.1) with c replaced by ηk/ω:

ED = πηku02 = 2πηE_So … (3.10.2)

In contrast, the energy dissipated in viscous damping increases linearly with the forcing frequency, as shown in Fig. 3.10.1.

Rate-independent damping is easily described if the excitation is harmonic and we are interested only in the steady-state response of this system. Difficulties arise in translating this damping mechanism back to the time domain. Thus it is most useful in the frequency-domain method of analysis (Section 1.10.3). The complex term k(1 + iη)u represents both the elastic and damping forces at the same time; k(1 + iη) is the complex stiffness of the system. There are advantages in this terminology if the reader is familiar with complex-variable notation. The complex stiffness has no physical meaning, however, in the same engineering sense as the elastic stiffness.

[Figure 3.10.1 - Energy dissipated in viscous damping and rate-independent damping. Plot of energy dissipated ED (vertical axis) versus forcing frequency ω (horizontal axis). A straight line through the origin, labeled “Viscous damping,” rises linearly with ω. A horizontal line, labeled “Rate-independent damping,” stays constant with ω. The two lines cross at ω = ωn. Image file: images/fig_3.10.1_energy_dissipated.png]

3.10.2 Steady-State Response to Harmonic Force

The equation for an SDF system with rate-independent linear damping, denoted by a crossed box in Fig. 3.10.2, is Eq. (3.2.1) with the damping term replaced by Eq. (3.10.1):

mü + (ηk/ω)u̇ + ku = p(t) … (3.10.3)

The mathematical solution of this equation is quite complex for arbitrary p(t). Here we consider only the steady-state motion due to a sinusoidal forcing function, p(t) = p0sin(ωt), which is described by

u(t) = u0 * sin(ωt - φ) … (3.10.4)

The amplitude u0 and phase angle φ are

u0 = (u_st)0 * 1 / sqrt( (1 - (ω/ωn)2)2 + η2 ) … (3.10.5)

φ = arctan( η / (1 - (ω/ωn)2) ) … (3.10.6)

These results are obtained by modifying the viscous damping ratio in Eqs. (3.2.11) and (3.2.12) to reflect the damping force associated with rate-independent damping, Eq. (3.10.1). In particular, $\zeta$ was replaced by

$\zeta = c/c_c = (\eta k/\omega) / (2m\omega n) = \eta / (2(\omega/\omega n))$ … (3.10.7)

Shown in Fig. 3.10.3 by solid lines are plots of u0/(u_st)0 and $\varphi$ as a function of the frequency ratio $\omega/\omega n$ for damping coefficient $\eta = 0$, 0.2, and 0.4; the dashed lines are described in the next section. Comparing these results with those in Fig. 3.2.6 for viscous damping, two differences are apparent: First, resonance (maximum amplitude) occurs at $\omega = \omega n$, not at $\omega < \omega n$. Second, the phase angle for $\omega = 0$ is $\varphi = \arctan(\eta)$ instead of zero for viscous damping; this implies that motion with rate-independent damping can never be in phase with the forcing function.

These differences between forced vibration with rate-independent damping and forced vibration with viscous damping are not significant, but they are the source of some difficulty in reconciling physical data. In most damped vibration, damping is not viscous, and to assume that it is without knowing its real physical characteristics is an assumption of some error. In the next section this error is shown to be small when the real damping is rate independent.

[Figure 3.10.2 - SDF system with rate-independent linear damping. Schematic: a mass m rests on a friction-free surface (shown with rollers), connected to a fixed wall by a spring of stiffness k and, in parallel, a crossed-box symbol labeled η (representing the rate-independent damping element). The mass displaces by u (arrow pointing right, labeled “u”) under applied force p(t) (arrow pointing right, labeled “p(t)”). Image file: images/fig_3.10.2_sdf_system.png]

[Figure 3.10.3 - Response of system with rate-independent damping: exact solution and approximate solution using equivalent viscous damping. Top plot: Deformation response factor Rd = u0/(u_st)0 (vertical axis, 0 to 5) versus frequency ratio ω/ωn (horizontal axis, 0 to 3). Solid curves show rate-independent damping for η = ζ = 0 (sharp peak at Rd = 5 at ω/ωn = 1), ζ = 0.1 with η = 0.2 (lower peak), and ζ = 0.2 with η = 0.4 (peak near Rd = 2.5). Dashed curves show the equivalent-viscous-damping approximation overlaid on each case. Bottom plot: Phase angle φ (vertical axis, 0° to 180°) versus the same frequency ratio ω/ωn. Curves rise from a small angle at ω/ωn = 0, pass through 90° at ω/ωn = 1, and approach 180° for large ω/ωn, again with solid (rate-independent) and dashed (equivalent viscous) curves for η = ζ = 0, η = 0.2/ζ = 0.1, and η = 0.4/ζ = 0.2. Image file: images/fig_3.10.3_response_factor.png]

3.10.3 Solution Using Equivalent Viscous Damping

In this section an approximate solution for the steady-state harmonic response of a system with rate-independent damping is obtained by modeling this damping mechanism as equivalent viscous damping.

Matching dissipated energies at $\omega = \omega n$ (Fig. 3.10.1) led to Eq. (3.9.2), where ED is given by Eq. (3.10.2), leading to the equivalent viscous damping ratio:

$\zeta_{eq} = \eta/2$ … (3.10.8)

Substituting this $\zeta_{eq}$ for $\zeta$ in Eqs. (3.2.10) to (3.2.12) gives the system response. The resulting amplitude u0 and phase angle $\varphi$ are shown by the dashed lines in Fig. 3.10.3. This approximate solution matches the exact result at $\omega = \omega n$ because that was the criterion used in selecting $\zeta_{eq}$. Over a wide range of excitation frequencies the approximate solution is seen to be accurate enough for many engineering applications. Thus Eq. (3.10.3) - which is difficult to solve for arbitrary force p(t) that contains many harmonic components of different frequencies $\omega$ - can be replaced by the simpler Eq. (3.2.1) for a system with equivalent viscous damping defined by Eq. (3.10.8). This is the basic advantage of equivalent viscous damping.

3.11 HARMONIC VIBRATION WITH COULOMB FRICTION

3.11.1 Equation of Motion

Shown in Fig. 3.11.1 is a mass-spring system with Coulomb friction force $F = \mu N$ that opposes sliding of the mass. As defined in Section 2.4, the coefficients of static and kinetic friction are assumed to be equal to $\mu$, and N is the normal force across the sliding surfaces. The equation of motion is obtained by including the exciting force in Eqs. (2.4.1) and (2.4.2) governing the free vibration of the system:

$m\ddot{u} + ku \pm F = p(t)$ … (3.11.1)

The sign of the friction force changes with the direction of motion; the positive sign applies if the motion is from left to right ($\dot{u} > 0$) and the negative sign is for motion from right to left ($\dot{u} < 0$). Each of the two differential equations is linear, but the overall problem is nonlinear because the governing equation changes every half-cycle of motion. Therefore, exact analytical solutions would not be possible except in special cases.

[Figure 3.11.1 - SDF system with Coulomb friction. Schematic: a mass m, connected to a fixed wall by a spring of stiffness k, rests on a surface exerting friction force ±F. The mass is subjected to applied force p(t) (arrow pointing right, labeled “p(t)”). Image file: images/fig_3.11.1_coulomb_friction_sdf.png]

3.11.2 Steady-State Response to Harmonic Force

An exact analytical solution for the steady-state response of the system of Fig. 3.11.1 subjected to harmonic force was developed by J. P. Den Hartog in 1933. The analysis is not included here, but his results are shown by solid lines in Fig. 3.11.2; the

dashed lines are described in the next section. The displacement amplitude u0, normalized relative to (u_st)0 = p0/k, and the phase angle φ are plotted as a function of the frequency ratio ω/ωn for three values of F/p0. If there is no friction, F = 0 and u0/(u_st)0 = (Rd)_ζ=0, the same as in Eq. (3.1.11) for an undamped system. The friction force reduces the displacement amplitude u0, with the reduction depending on the frequency ratio ω/ωn.

[Figure 3.11.2 - Deformation response factor and phase angle of a system with Coulomb friction excited by harmonic force. Exact solution from J. P. Den Hartog; approximate solution is based on equivalent viscous damping. Top plot: Deformation response factor Rd = u0/(u_st)0 (vertical axis, 0 to 3.5) versus frequency ratio ω/ωn (horizontal axis, 0 to 2.0). Solid curves (Coulomb friction) are shown for F/p0 = 0, 0.5, 0.7, π/4, and 0.8, each rising to a peak near ω/ωn = 1 and diverging to infinity at ω/ωn = 1 for the lower friction values; the F/p0 = 0.8 curve shows a rounded peak around Rd ≈ 1.6 near ω/ωn ≈ 0.85 and then drops off. Dashed curves (equivalent viscous damping) approximate each solid curve. Bottom plot: Phase angle φ (vertical axis, 0° to 180°) versus the same frequency ratio. Curves for F/p0 = 0 (φ = 0 for ω/ωn < 1, jumping to ~180° above resonance), and for F/p0 = 0.5, 0.7, 0.8, with corresponding dashed equivalent-viscous-damping approximations. Image file: images/fig_3.11.2_coulomb_response.png]

## 3.1 | section 1

This example illustrates arrays and plotting. [4]

CLOUGH, PENZIEN - Dynamics of Structures, page 178

Use flexibility formulation (see page 182)

calioPY Procedure Output - Example 6 This example illustrates arrays and plotting. =CLOUGH, PENZIEN - Dynamics of Structures, page 178 Use flexibility formulation (see page 182)

1. set up mass and stiffness arrays m = PL.array([[1.0,0,0],[0,1.5,0],[0,0,2.0]],float) [1] k1 = 600·PL.array([[1,-1,0.0],[-1,3,-2],[0,-2,5]],float) [2]

2. flexibility and dynamic matrix f = PL.inv(k1) [3] d = PL.inner(f,m) [4]

3. eigenvalues eigen = PL.eig(d) [5] evalu = eigen[0] [6] . evalu = [ 0.00474 0.00104 0.00047] nat_freq = 1/(evalu^.5) [7] . nat_freq = [ 14.52482 31.00868 46.12656]

4. normalize and scale eigenvectors evect = PL.array(eigen[1]) [8] . evect = [[-0.813 -0.739 0.273] [-0.527 0.449 -0.694] [-0.246 0.502 0.666]]

[py] for i in range(len(nat_freq)): [py] evectt = PL.transpose(evect) [py] evectt[i] = evectt[i]/evectt[i][0]

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## 3.1 - 2 | section-2

normalized eigenvectors

[[ 1. 0.648 0.303] [ 1. -0.608 -0.679] [ 1. -2.542 2.44 ]] SUMMARY TABLES

mass matrix 1.0 0.0 0.0 0.0 1.5 0.0 0.0 0.0 2.0

stiffness matrix 600.0 -600.0 0.0 -600.0 1800.0 -1200.0 0.0 -1200.0 3000.0

flexibility matrix 0.0031 0.0014 0.0006 0.0014 0.0014 0.0006 0.0006 0.0006 0.0006

dynamic matrix 0.003 0.002 0.001 0.001 0.002 0.001 0.001 0.001 0.001

xx = NP.concatenate((evals,evect),1) yy = [“freq”,”level 3”,”level 2”,”level 1”] tt = NP.vstack((yy,xx)) . tt = [[‘freq’ ‘level 3’ ‘level 2’ ‘level [‘14.52482’ ‘1.0’ ‘1.0’ ‘1.0’] [‘31.00868’ ‘0.648’ ‘-0.608’ ‘-2.542’] [‘46.12656’ ‘0.303’ ‘-0.679’ ‘2.44’]]

eigenvectors and eigenvalues freq level 3 level 2 level 1 14.52482 1.0 1.0 1.0 31.00868 0.648 -0.608 -2.542 46.12656 0.303 -0.679 2.44

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## 3.1 - 3 | Plot Mode Shapes

1. set up mass and stiffness arrays [1] m = PL.array([[1.0,0,0],[0,1.5,0],[0,0,2.0]],float [2] k1 = 600PL.array([[1,-1,0.0],[-1,3,-2],[0,-2,5]

2. flexibility and dynamic matrix [3] f = PL.inv(k1) [4] d = PL.inner(f,m)

3. eigenvalues [5] eigen = PL.eig(d) [6] evalu = eigen[0] ?5 [7] nat_freq = 1/(evalu.5) ?5

4. normalize and scale eigenvectors [8] evect = PL.array(eigen[1]) ?

-pycode- for i in range(len(nat_freq)): evectt = PL.transpose(evect) evectt[i] = evectt[i]/evectt[i][0] -insert-

normalized eigenvectors {} evectt ?3

SUMMARY TABLES

-table-02 m ?4; mass matrix;

-table-02 k1 ?4; stiffness matrix;

-table-02 f ?4; flexibility matrix ;

-table-02 d ?3; dynamic matrix ;

-k- # use reshape to transpose 1d array (list)

evals=PL.reshape(nat_freq, (len(nat_freq),1)) xx = NP.concatenate((evals,evect),1) yy = ["freq","level 3","level 2","level 1"] tt = NP.vstack((yy,xx)) ? -table-01 tt ?3; eigenvectors and eigenvalues; . # initialize eigenvector array (need (x,1) shapes for plotting ms = PL.shape(evect) zz = PL.zeros((ms[0],1)) x1=PL.concatenate((evect,zz),1) -pycode- #plot mode shapes using pylab y=PL.array([0,1,2,3]) m3=x1[2].35+3 m2=x1[1].35+2 m1=x1[0].35+1 m=PL.concatenate((m1,m2,m3)) PL.clf() PL.plot(m1,y) PL.plot(m2,y) PL.plot(m3,y) PL.xlim(.5,4.) PL.xlabel('mode') PL.ylabel('levels') PL.title("Mode Shapes ghPenzien (page 178)") PL.grid() PL.savefig(_cpypath+"/figs/p178a.png")

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1 P.B. Lindley, Engineering Design with Natural Rubber, NR Technical Bulletin.Malaysian Rubber Producers' Research Association, Brickendonbury, U.K.

2 Anil K.Anil K. Chopra, Dynamics of Structures: Theory and Applications toEarthquake Engineering. Englewood Cliffs, NJ, USA: Prentice Hall, 1995.

3 Anil K.Anil K. Chopra, Dynamics of Structures: Theory and Applications toEarthquake Engineering. Englewood Cliffs, NJ, USA: Prentice Hall, 1995.

4 R.W. Clough and J. Penzien, *Dynamics of Structures*. New York, NY, USA:McGraw-Hill, 1975.